Lecture 13: Exoplanets
Detection, Demographics, Characterisation

Planetary Systems

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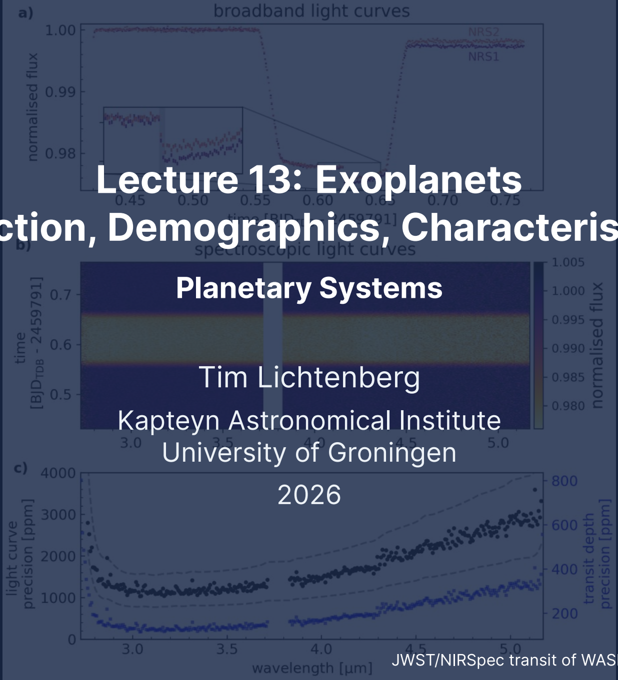


JWST/NIRSpec transit of WASP-39 b. Credit: Alderson et al. (2023)

Learning Objectives

By the end of this lecture, you will be able to:

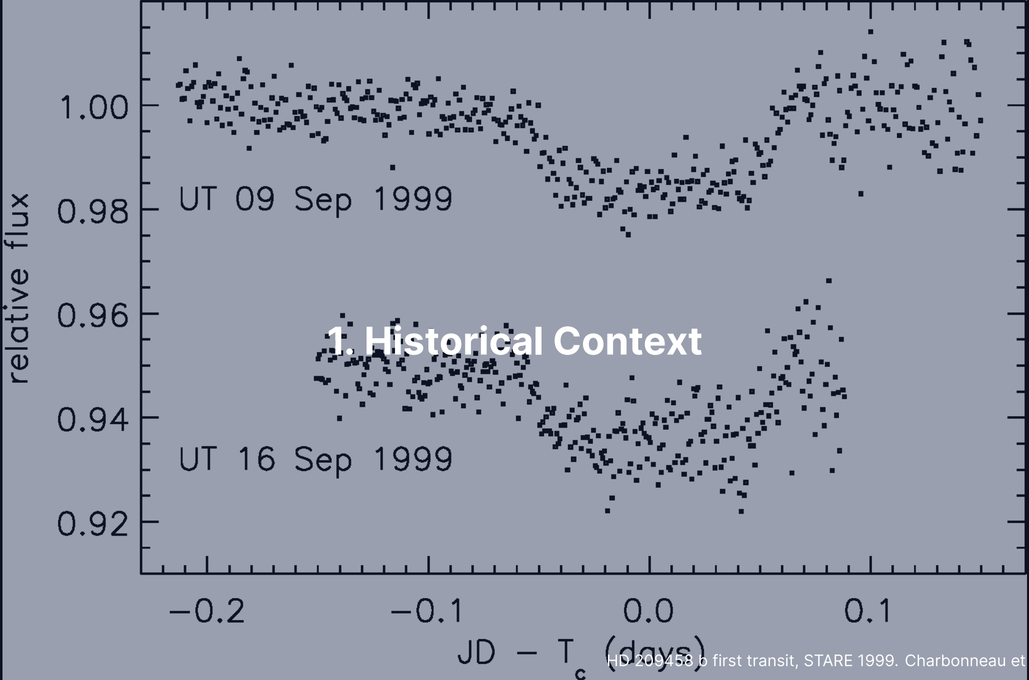
- Describe the main exoplanet detection methods and their biases.
- Interpret the period-radius diagram and the radius valley.
- Apply transit + radial-velocity geometry to derive mass, radius, and bulk density.
- Evaluate JWST-era atmospheric results and biosignature claims.

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

- L9-L11 walked through every major solar system planet.
- L12: small bodies as formation fossils.
- Today: *turn the question outward*. Are these patterns universal?
- L14: synthesis. What constitutes a convincing detection of life?

1. Historical Context

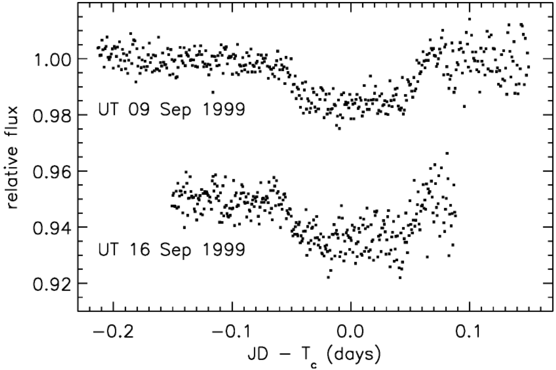


HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

From Zero to Six Thousand in Three Decades

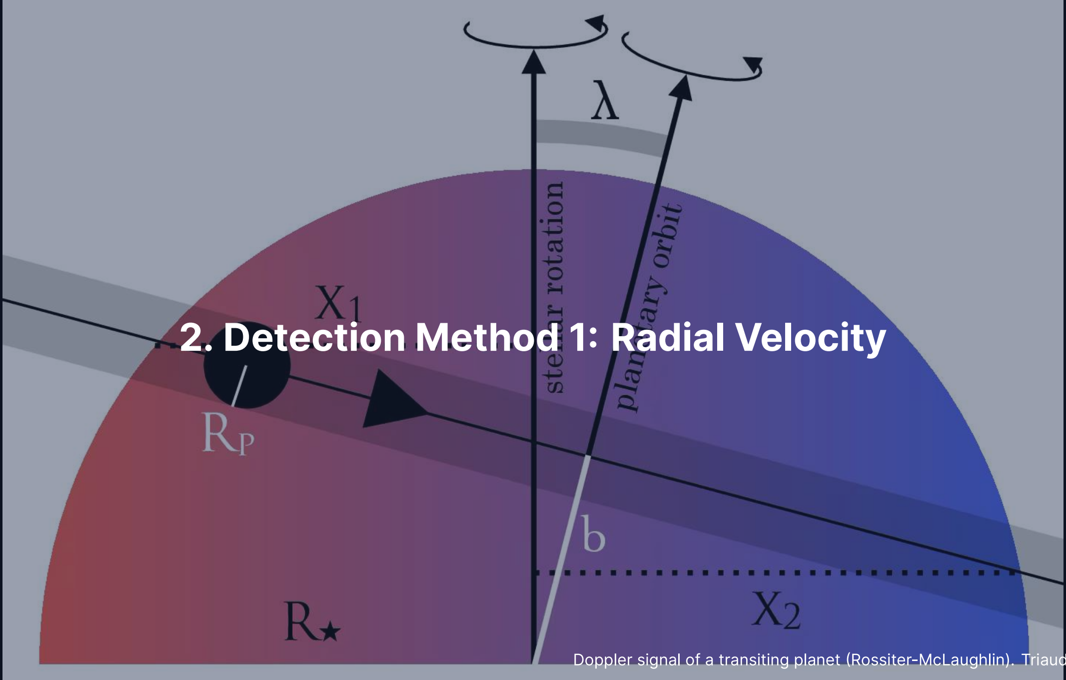
- **1992:** Wolszczan + Frail. Two planets around pulsar PSR B1257+12 from pulse-timing.
- **1995:** Mayor + Queloz. **51 Pegasi b** via radial velocity. 0.5 M_{J}, 4.23-day orbit.
  - Contradicted every textbook picture of giant-planet formation.
  - → migration models reshaped formation theory.
  - Nobel Prize 2019.
- **1999:** HD 209458 b. First transit of a known RV planet.
  - Combined transit depth + RV → first bulk density of an exoplanet.
  - Confirmed hot Jupiters are gas-dominated.
- **2009–2018:** Kepler + K2 transformed the field statistically.
- **2022–present:** JWST atmospheric characterisation era.

HD 209458 b: First Ground-Based Transit (1999)



- STARE photometer, two nights in Sep 1999.
- Successive transits dropped flux by ~ 1.7%.
- Independent detection by Henry et al. (2000).
- After this point: exoplanets were physical objects, not just Doppler signals.

2. Detection Method 1: Radial Velocity



Doppler signal of a transiting planet (Rossiter-McLaughlin). Triaud (2018)

Radial Velocity: Doppler Reflex of the Star

- Star + planet orbit a common barycentre (centre of mass).
- Star's reflex orbit: $a_\star = (m_p/M_\star)\,a_p$.
- Spectral lines blueshift / redshift periodically.
- For circular, edge-on orbit (semi-amplitude $K_\star$, the peak reflex velocity):

$$K_\star = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}} \frac{1}{\sqrt{1-e^2}}$$

Quick scales:

- Jupiter at 5.2 AU: $K_\star \approx 12.5\ \mathrm{m\ s^{-1}}$.
- Earth at 1 AU: $K_\star \approx 9\ \mathrm{cm\ s^{-1}}$.

Instrument Floor and Stellar Noise

- ELODIE (1995): ~10 m s^{-1}.
- HARPS (2003): ~1 m s^{-1}.
- ESPRESSO (2018): 10 cm s^{-1} demonstrated.
- Now **stellar noise**-limited: granulation, p-modes, starspots produce 10 cm s^{-1} to 1 m s^{-1} jitter even on quiet stars.
- True Earth twin: at the very edge of physical accessibility.

The $m \sin i$ Degeneracy

RV gives only $K_\star$, hence $m_p \sin i$.

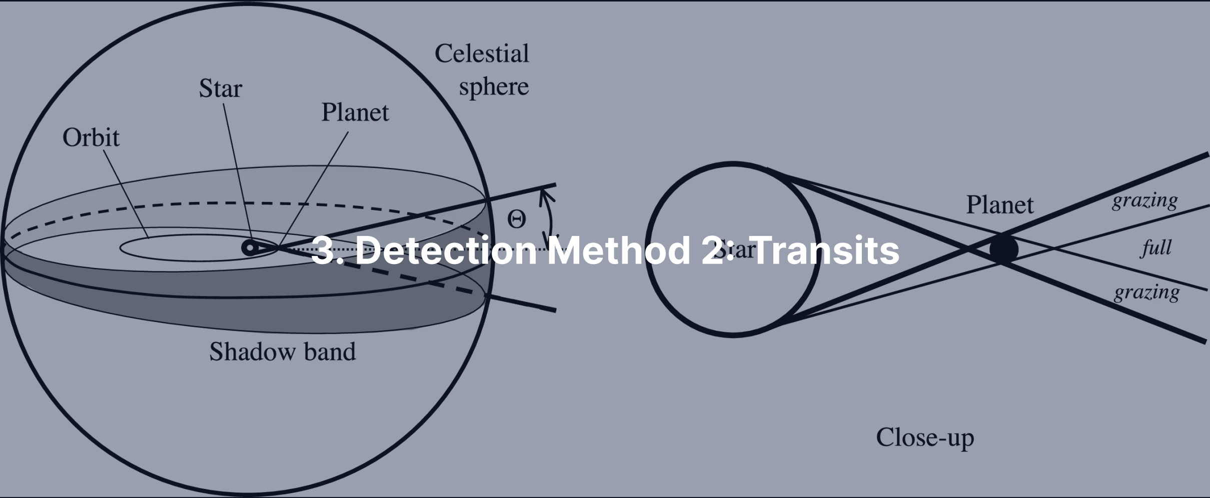
- Face-on: $\sin i = 0$, no signal at all.
- Edge-on: $\sin i = 1$, true mass.
- RV alone → *minimum* mass only.

Break the degeneracy with:

- **Transit** ($\sin i \approx 1$ to a few percent),
- **Astrometry** (gives inclination directly),
- **Direct imaging** (geometric).

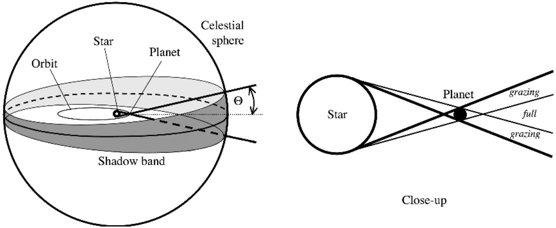
RV biases: massive planets, short orbits, bright slowly-rotating quiet G/K stars.

3. Detection Method 2: Transits



Transit geometry. Winn (2010)

Transit Geometry



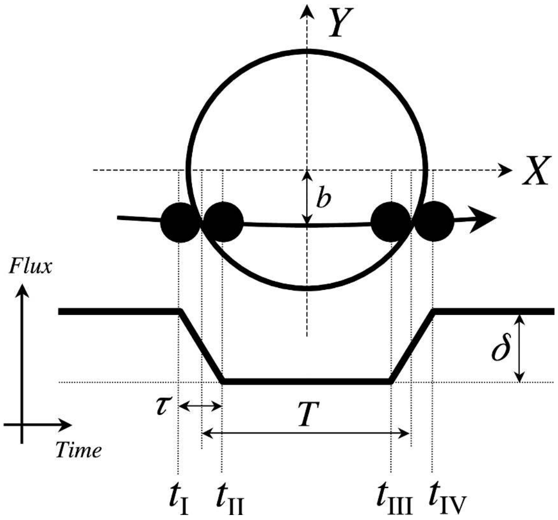
- Orbital plane near our line of sight → planet crosses stellar disk.
- Depth: $\delta = (R_p/R_\star)^2$.
- Probability: $\Theta \approx R_\star/a$.
- Strong selection: short periods + large $R_p/R_\star$.

Quick Scales

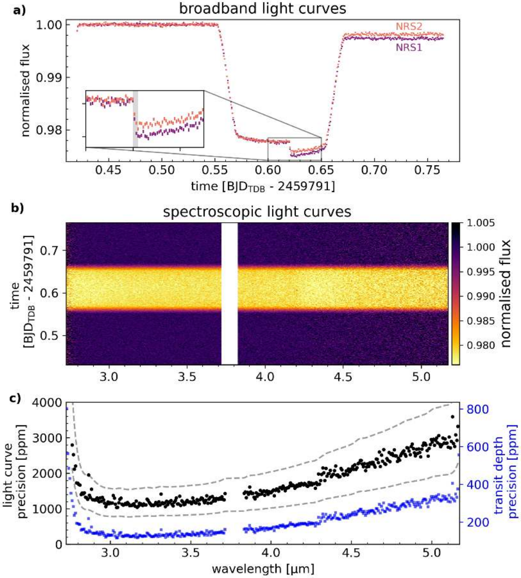
- Jupiter (R_J) around Sun ($R_\odot$): $\delta \approx 1.01\%$. Easy from the ground.
- Earth around Sun: $\delta \approx 84$ ppm. Requires space.
- Earth around M dwarf (small, cool star; $R_\star = 0.15\,R_\odot$): $\delta \approx 4000$ ppm. Ground-accessible!

Inverse scaling with stellar radius is why TRAPPIST-1 dominates atmospheric characterisation.

Light Curve Anatomy



- Four contact times: t_{I}, t_{II}, t_{III}, t_{IV}.
- Duration → impact parameter b (transit offset from disk centre), inclination.
- Ingress shape → stellar limb darkening.
- Depth → planet radius (given $R_\star$).

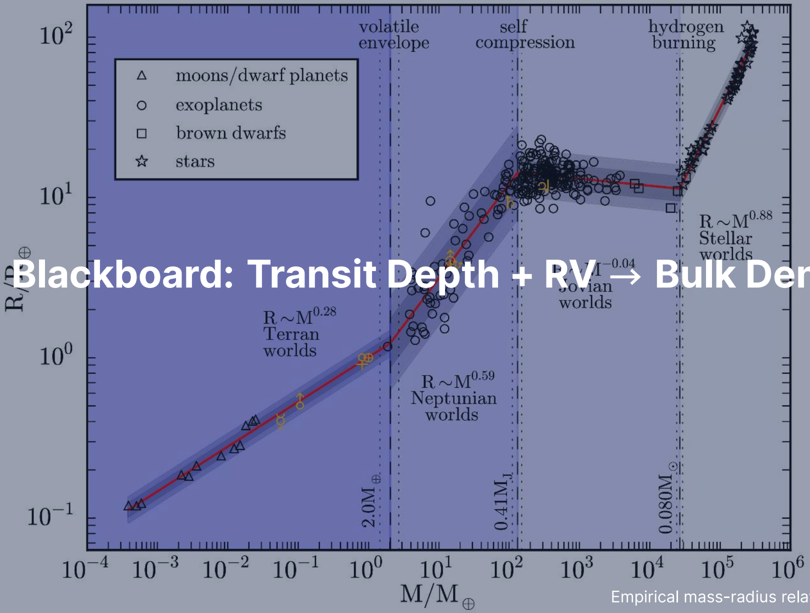


JWST/NIRSpec transit photometry of WASP-39 b: transit depth measured to 200–600 ppm per spectroscopic bin. Alderson et al. (2023).

Reading the JWST Transit Photometry

- **Top**: raw broadband transit light curves, NIRSpec G395H NRS1 + NRS2.
- **Middle**: spectroscopic light curves at each wavelength bin.
- **Bottom**: per-channel precision → 200–500 ppm per integration.
- Photometric noise floor is now *below* typical exoplanet absorption features.
- Selection biases:
  - Short period ($P < 100$ days) for repeated transits.
  - Large $R_p/R_\star$ (favours hot Jupiters and M-dwarf systems).

4. Blackboard: Transit Depth + RV → Bulk Density



Empirical mass-radius relation. Chen & Kipping (2017)

Why Combine Transit and RV?

- Transit alone: R_p, but no mass.
- RV alone: $m_p \sin i$, but no radius.
- Combined: **both** for the same planet → bulk density.
- Density is the central observable that turns a detection into a physical object.

Step 1: Transit Depth

Star = uniform disk $R_\star$. Planet = opaque disk R_p.

$$\frac{\Delta F}{F} = \frac{\pi R_p^2}{\pi R_\star^2} = \left(\frac{R_p}{R_\star}\right)^2$$

Given $R_\star$ from spectroscopy + parallax, the depth gives R_p directly.

Limb darkening + ingress shape → ~ 10% corrections.

Step 2: RV Semi-Amplitude

Conservation of momentum: $M_\star a_\star = m_p a_p$, $a = a_\star + a_p$.

$$v_\star = \frac{2\pi a_\star}{P} = \frac{m_p}{M_\star + m_p} \cdot \frac{2\pi a}{P}$$

Use Kepler: $a^3 = G(M_\star + m_p)P^2/(4\pi^2)$, sub in:

$$K_\star = v_\star \sin i = \left(\frac{2\pi G}{P}\right)^{1/3} \frac{m_p \sin i}{M_\star^{2/3}} \frac{1}{\sqrt{1-e^2}}$$

$K_\star \propto P^{-1/3}$: short-period planets give larger reflex.

Step 3: Combine

A transit forces $\sin i \approx 1$ (orbit near edge-on).

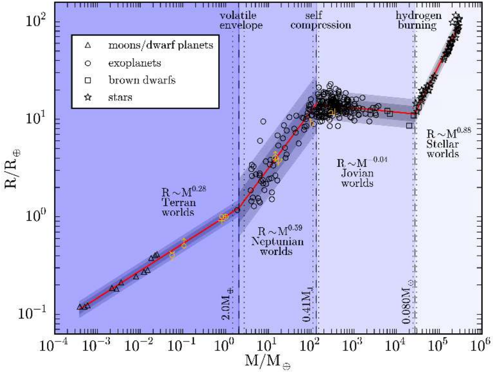
- $\sin i \approx 1$ collapses the $m_p \sin i$ degeneracy.
- Both R_p and m_p measured for the same object.
- Bulk density follows from elementary geometry:

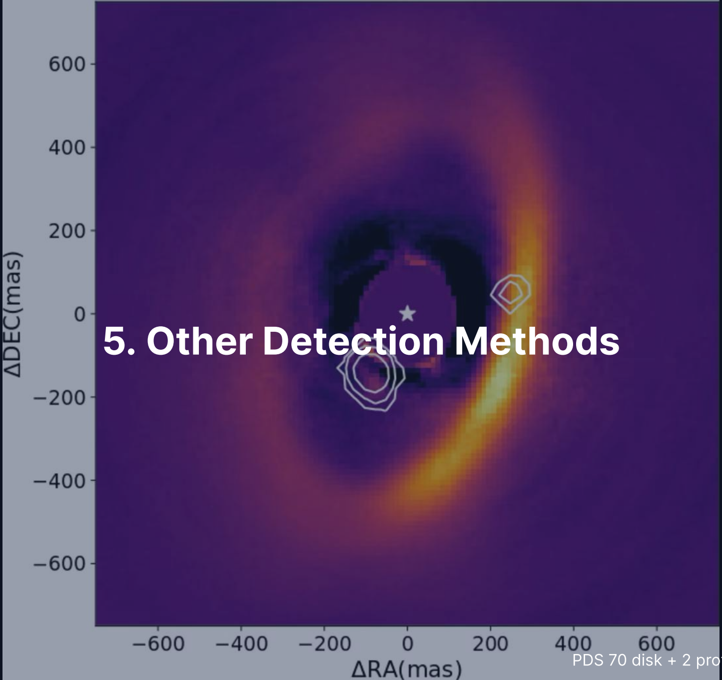
$$\boxed{\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}}$$

Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.

Density Maps to Composition

- $\bar{\rho} \approx 5.5$: rocky (Earth, Venus).
- $\bar{\rho} \approx 1.3$: gas giant (Jupiter).
- $\bar{\rho} \approx 0.5$: inflated H/He envelope.
- $\bar{\rho} \approx 2$–4: *sub-Neptune*, the most common type with no solar-system analogue.





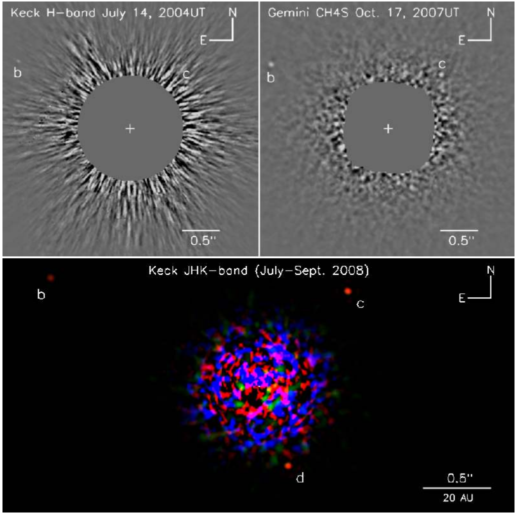
5. Other Detection Methods

PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

Direct Imaging: Spatial Separation of Photons

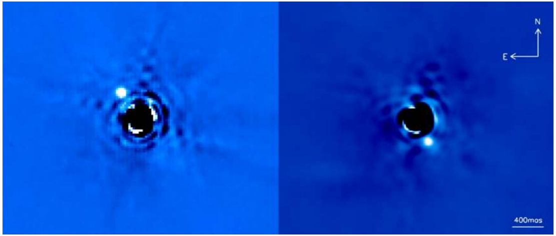
- Conceptually simplest, technically hardest.
- Sun + Jupiter contrast at visible: 10^{-9}. Sun + Earth: 10^{-10}.
- Four ingredients:
  - Adaptive optics (atmospheric turbulence correction)
  - Coronagraphs (stellar light suppression)
  - Angular + spectral differential imaging (speckle subtraction)
  - Long integration (→ averaging speckle realisations)
- Strongly biased to *young*, massive, wide-orbit, self-luminous giants.

HR 8799: Four Imaged Giants

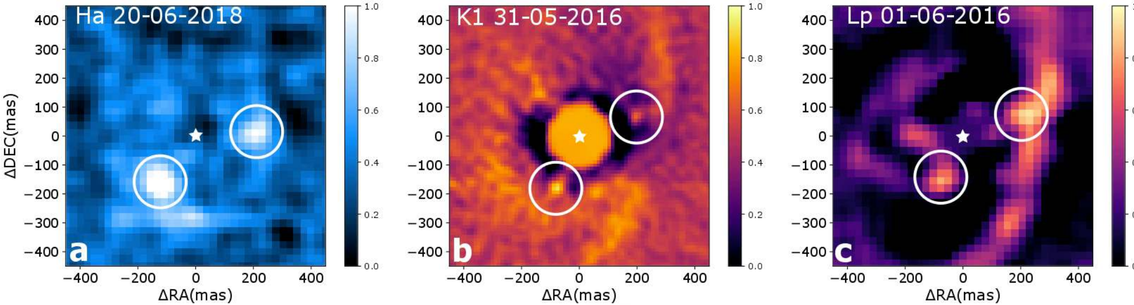


- Marois et al. 2008. Keck + Gemini AO; sub-panels show three discovery epochs (HK 2004, CH_4 2007, JKH 2008) confirming bound co-moving companions.
- Three planets initially, fourth added 2010.
- Each ~ 5–10 M_J at 14–70 AU.
- Host: 30 Myr A-type. Planets still self-luminous and cooling.

β Pictoris b: Orbital Motion in Two Epochs

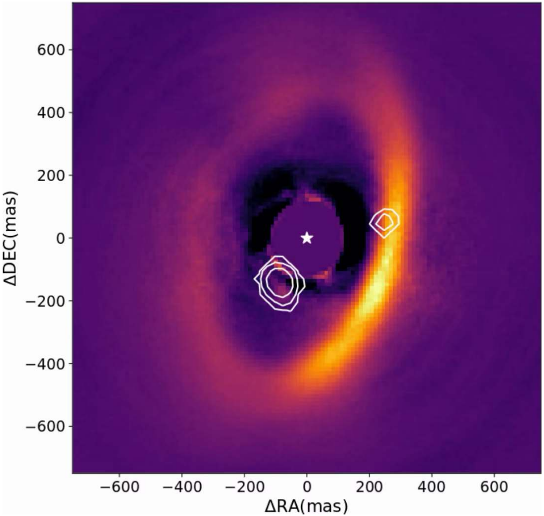


- Lagrange et al. (2010).
- ~ $8\,M_{\mathrm{J}}$ at ~ 9 AU.
- Two epochs (2003, 2009) → orbital motion confirms bound companion.
- Same star: imaged debris disk (leftover dust ring) + a second giant (c).



PDS 70 b + c inside the disk gap in three bands: MUSE Hα (a), SPHERE K1 (b), NACO L′ (c). Haffert et al. (2019).

PDS 70: Forming Planets in a Disk Gap



- Host: 5 Myr K7 with cleared central cavity.
- SPHERE (NIR) + MUSE (Hα) detected b inside the gap (Keppler+ 2018).
- Second protoplanet c added later (Haffert+ 2019).
- Both are *actively accreting* (Hα shock emission).
- Cleanest direct match between planet-disk theory and observation.

Astrometry: Gaia Will Find Jupiter Analogues

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- Jupiter analogue at 10 pc: ~ 0.5 mas. Earth analogue: $\sim 0.3\,\mu$as.
- Hipparcos (1989–93): ~ 1 mas, no exoplanets.
- Gaia (2013–): 20–50 μas per epoch, $\sim 10\,\mu$as final.
- DR4 (late 2026): 10^2–10^3 astrometric exoplanets, mostly Jupiter analogues.
- Astrometry gives *inclination directly* → true mass.

Microlensing + Timing

- **Microlensing**: foreground star magnifies background source; planet adds a small spike.
  - Sensitive at kpc distances; sweet spot at 0.5–10 AU.
  - One-shot, never repeats.
  - Roman (2027): expected ~ 1000–3000 bound + comparable rogue planets.
- **TTVs**: gravitational coupling between transiting planets perturbs transit times → dynamical masses without RV.
  - Essential for faint stars: TRAPPIST-1 masses come almost entirely from TTVs.
- **Pulsar timing**: extraordinarily sensitive but tiny target sample.
- **Eclipsing binaries**: Kepler-16 b ("Tatooine") is circumbinary (orbits both stars).

Detection Bias Summary

- **RV**: massive, 1–10 yr orbits, bright G/K stars.
- **Transit**: short period, large $R_p / R_\star$, quiet bright stars.
- **Direct imaging**: wide (> 10 AU), young, massive, self-luminous.
- **Astrometry**: wide orbits matching mission baseline.
- **Microlensing**: 1–10 AU, any host distance, unrepeatable.
- **Timing**: compact multis (TTVs) or circumbinaries.

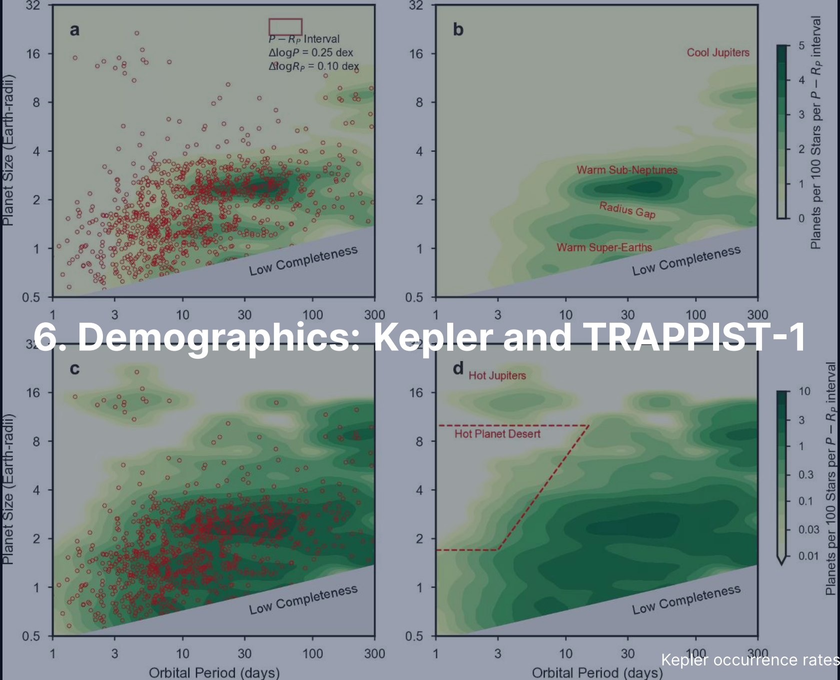
The shape of the observed exoplanet archive reflects the union of biases as much as the underlying physics.

Break

End of Part 1: Detection Methods

Part 2: Demographics, Architectures, Atmospheres

6. Demographics: Kepler and TRAPPIST-1

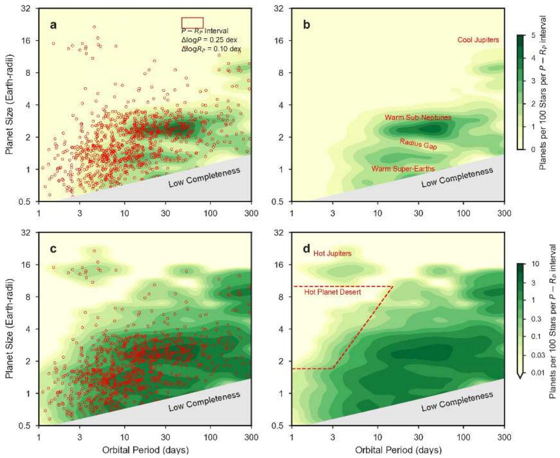


Kepler occurrence rates. Petigura et al. (2018)

The Kepler Revolution

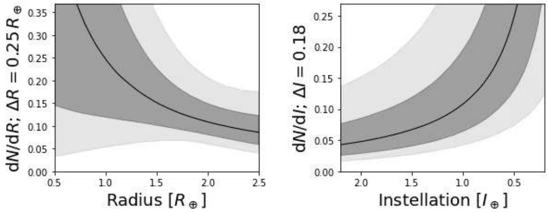
- 0.95 m telescope, single field, 150,000 stars, 4 years, 30 ppm precision.
- *Computable detection efficiency* → first occurrence-rate measurements (planets per star, bias-corrected).
- Headline: at least one planet per main-sequence star (Petigura+ 2018).
- Small planets ($R < 4\,R_\oplus$) dominate by far.
- Hot Jupiters: 0.5–1% of Sun-like stars (rare, but easy to find).
- $\eta_\oplus$ (Earth-size in HZ, the habitable zone): ~ 0.1–0.6, central value ~ 0.3 (Bryson+ 2021).

Period-Size Occurrence Rate



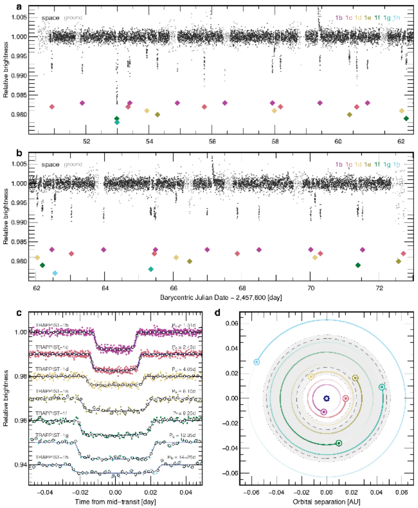
- California-Kepler Survey, after stellar parameter refinement (sub-panels label distinct period-radius regimes).
- Small planets common at every period.
- Slight fall-off at longer periods (partly observational).
- Long-period flattening → Kepler had only 4-yr baseline.

$\eta_\oplus$: Earths in Habitable Zones



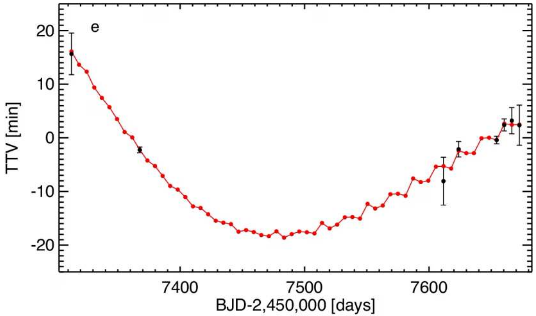
- Bryson+ 2021 marginal occurrence.
- Left: per unit radius. Right: per unit instellation (stellar flux received).
- Dark/light bands: 68% / 95% credible intervals.
- $\eta_\oplus$: 0.1–0.6 depending on definition.
- **Earth-class planets are ordinary, not miraculous.**

TRAPPIST-1: Seven Earths in Resonance



- M8 ultra-cool dwarf at 12 pc.
- Seven transiting Earth-sized planets within 0.06 AU.
- Five-step mean-motion resonant chain (period ratios near small integers) → early disk migration.
- Probability all seven transit if random: $< 10^{-3}$.
- Gillon et al. 2017.

TRAPPIST-1 Masses from TTVs

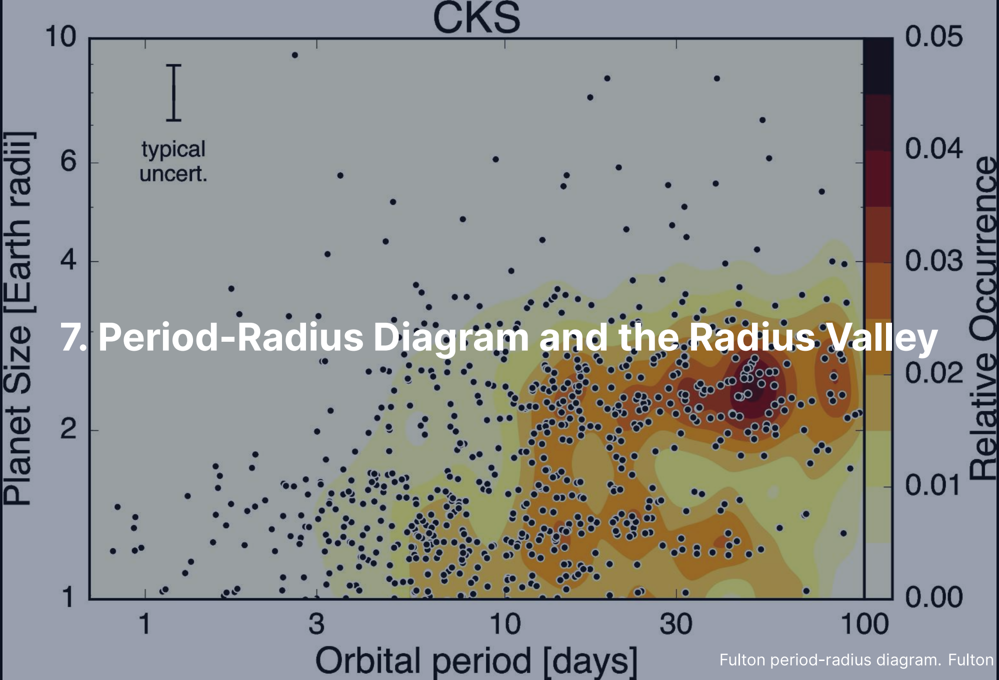


- TRAPPIST-1 e transit-timing variations.
- Tens-of-minutes amplitude over hundreds of days.
- Best-fit dynamical model includes the full 7-planet coupling.
- Yields all seven masses without RV.
- M8 is too faint for high-precision RV.

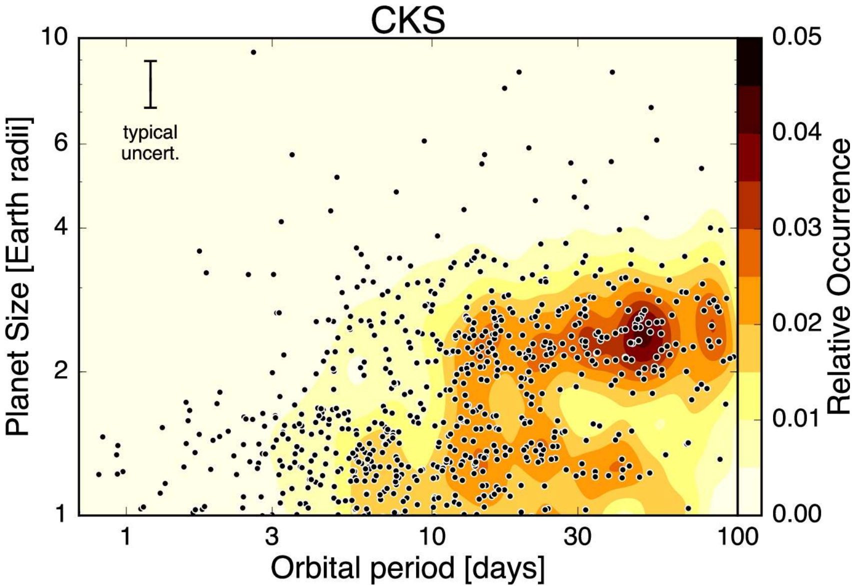
Break

Earthrise, Apollo 8 (1968). Credit: NASA

7. Period-Radius Diagram and the Radius Valley



Fulton period-radius diagram. Fulton et al. (2017)

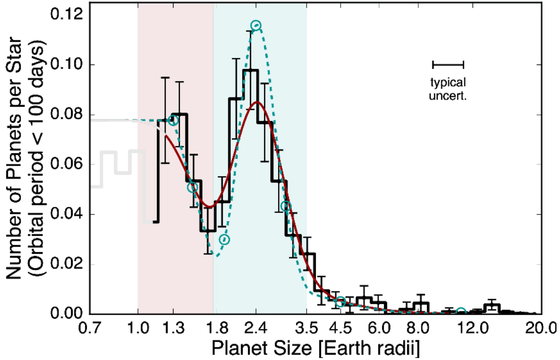


Period-radius distribution of small Kepler planets after stellar parameter refinement. Fulton et al. (2017).

Reading the Period-Radius Diagram

- Hot Jupiters: tight clump at $P < 10$ d, $R > 10\,R_\oplus$.
- Sub-Neptunes (2–4 $R_\oplus$): *the most common class*; no solar-system analogue.
- Super-Earths (1–1.8 $R_\oplus$): rocky planets larger than Earth.
- Terrestrial regime ($P > 100$ d, $R < 1.5\,R_\oplus$): largely unexplored. PLATO will fill.
- Most striking feature: the **deficit at** $\sim 1.8\,R_\oplus$.

The Radius Valley (Fulton Gap)



- Statistically significant deficit at 1.5–2 $R_\oplus$.
- Below: super-Earths peaked at ~ 1.3 $R_\oplus$.
- Above: sub-Neptunes peaked at ~ 2.4 $R_\oplus$.
- Bimodal small-planet population. Defining demographic feature.

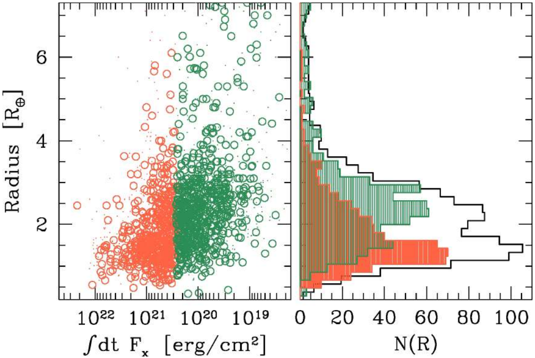
Photoevaporation Strips H/He Envelopes

Energy-limited escape rate:

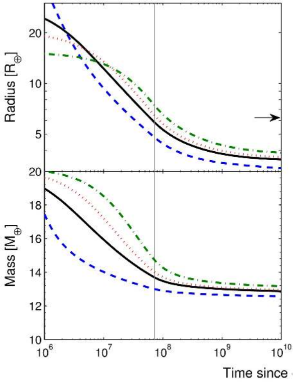
$$\dot{M} \approx \frac{\epsilon \pi F_{\mathrm{XUV}} R_p^3}{G M_p}$$

- Young stars saturate XUV at $\sim 10^3 \times$ present-day for ~ 100 Myr.
- Low-density, low-mass planets are *easier to strip*.
- Cumulative loss for a $10\,M_\oplus$ sub-Neptune at 0.1 AU: a few % of an Earth mass.
- Comparable to a primordial H/He envelope on a rocky core.
- Owen & Wu (2013): naturally produces a bimodal radius distribution.

Photoevaporation Model + Mass-Loss Tracks

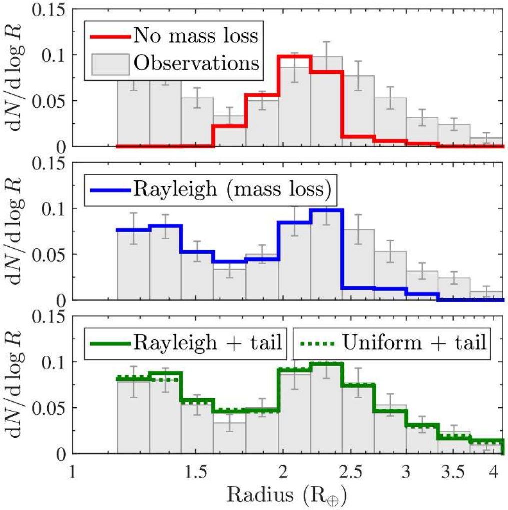


Left: predicted radius valley



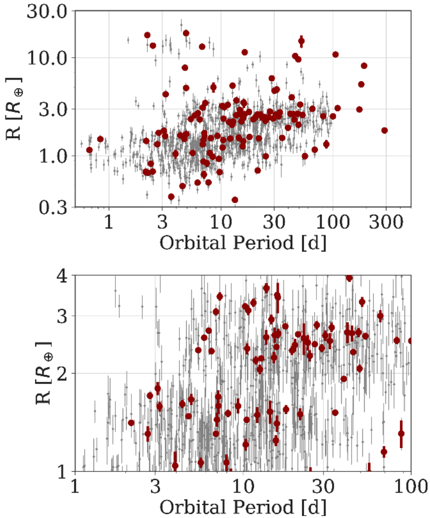
Right: time evolution of radius + mass; saturated XUV phase out to ~ 100 Myr drives bimodal final distribution.

Core-Powered Mass Loss: Same Outcome, Different Driver



- Ginzburg+ 2018, Gupta+ 2019.
- Hot rocky core retains formation heat → basal flux into envelope drives slow outflow.
- No XUV needed.
- **Bottom panel** (Rayleigh + tail): bimodal radius distribution at ~ 1.8 $R_\oplus$, mirroring photoevaporation.
- Both mechanisms likely operate.

Slope of the Valley with Period

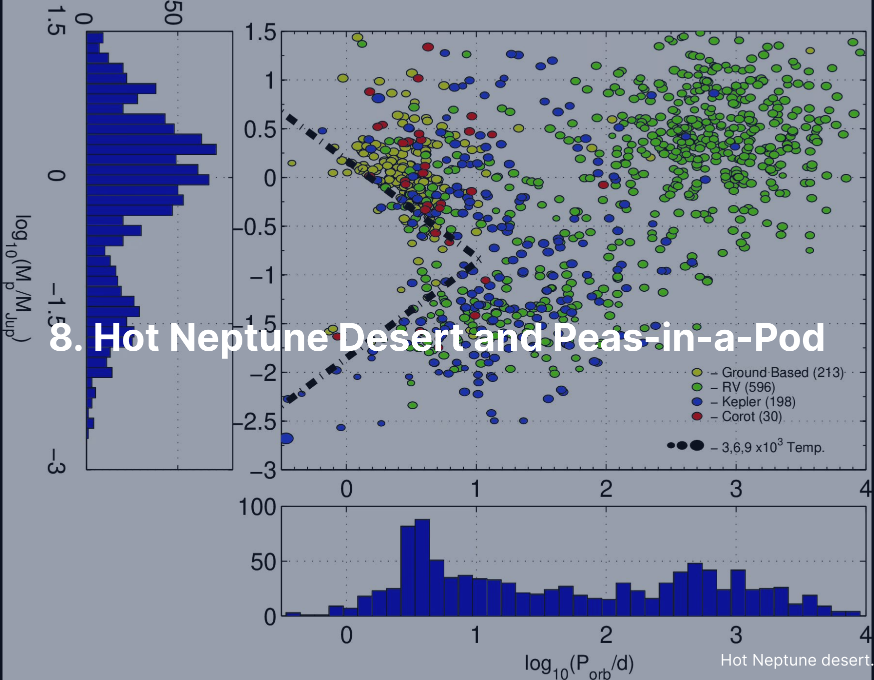


- Van Eylen+ 2018, asteroseismic radii (from stellar oscillations).
- Valley shifts to smaller R_p at longer periods.
- Brown: super-Earths. Blue: sub-Neptunes.
- Slope consistent with both photoevaporation and core-powered models.
- Current data cannot distinguish.

Implication: Many Super-Earths Are Stripped Cores

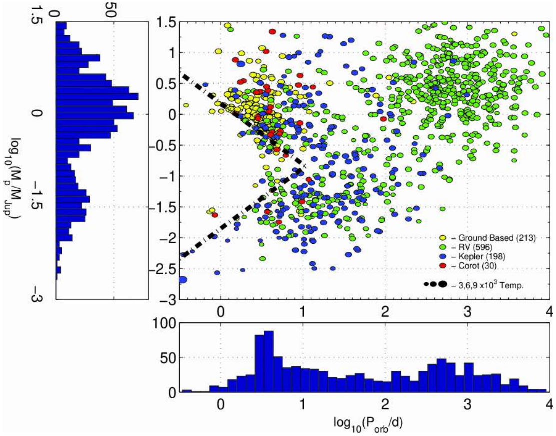
- Many close-in rocky super-Earths are not ”naturally” rocky.
- They are the bare cores of former sub-Neptunes.
- Whether a given super-Earth was born rocky or stripped: *cannot tell from bulk density alone*.
- Consequence for habitability: stripped cores have violent atmospheric histories and may have very different surface/interior chemistry.

8. Hot Neptune Desert and Peas-in-a-Pod



Hot Neptune desert. Mazeh et al. (2016)

The Hot Neptune Desert

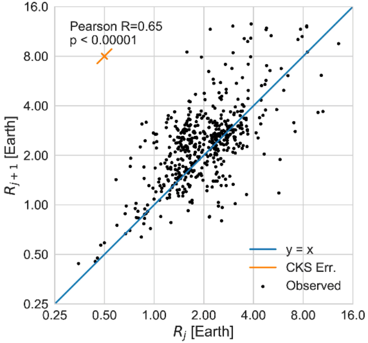


- Striking deficit at 10–100 $M_\oplus$, $P < 5$ d.
- Hot Jupiters at the same periods: rare but present.
- Ultra-short rocky: rare but present.
- Hot Neptunes: *essentially absent*.
- Power-law boundaries (Mazeh+ 2016).

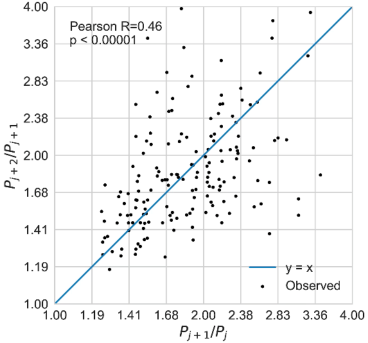
Two Edges, Two Mechanisms

- **Lower edge (small mass)**: photoevaporation strips weakly-bound Neptune envelopes → leaves super-Earths.
- **Upper edge (large mass)**: Roche-lobe overflow (tidal gas stripping) on inflated hot Jupiters truncates the heavy population.
- Independent evidence that mass loss is the dominant sculptor of close-in planets.

Peas-in-a-Pod



Adjacent radii: Pearson 0.65



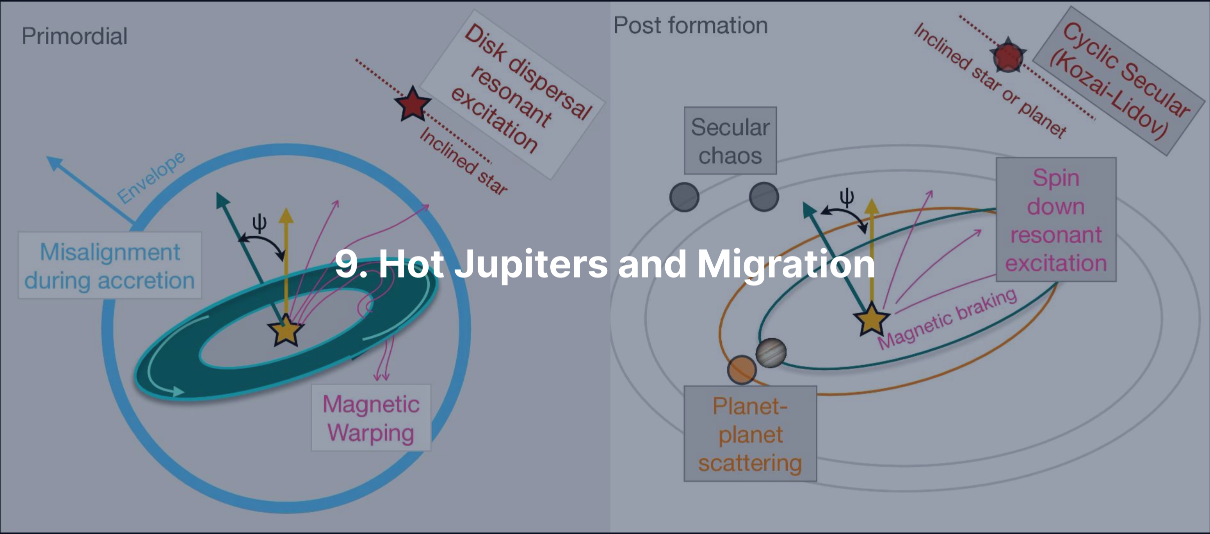
Adjacent period ratios: clustered

Within a Kepler multi-system, planets tend to be *the same size* as each other and *uniformly spaced*. Weiss et al. (2018).

Peas-in-a-Pod: Implications

- Implies smooth, local formation rather than stochastic giant impacts.
- Inner solar system (giant-impact dominated) may be unusual.
- TRAPPIST-1: extreme example of a peas-in-a-pod with resonant chain.
- Kepler-90 (8 planets), TOI-178 (6 planets in unbroken chain).
- Not universal: RV-discovered systems show much more architectural diversity.

9. Hot Jupiters and Migration



Hot Jupiter migration channels. Albrecht et al. (2022)

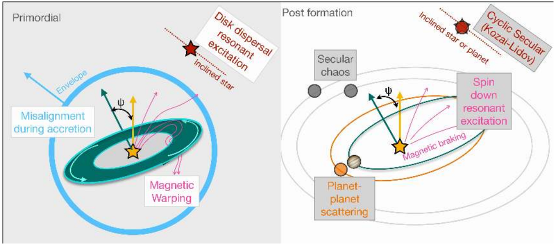
Three Migration Channels

Hot Jupiters at 0.05 AU *must* have migrated.

- **Disk migration** (Type II): smooth, gas-driven. Preserves alignment.
- **High-eccentricity migration** (Kozai-Lidov): distant perturber pumps eccentricity, tides circularise. Produces misalignments.
- **Planet-planet scattering**: dynamical instability ejects giants, leaves survivors on eccentric orbits, tides circularise. Misalignments.

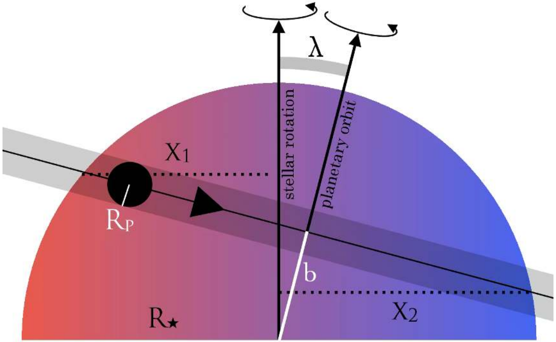
Inner edge set by Roche limit + tidal circularisation timescale: pile-up at ~ 0.04–0.05 AU.

Migration Channels and Stellar Obliquities



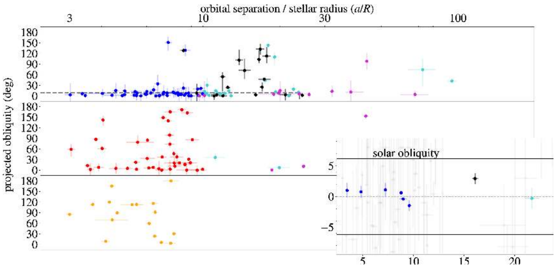
- Albrecht+ 2022.
- Disk migration → aligned.
- Kozai-Lidov → arbitrary obliquity.
- Scattering → misaligned.
- Stellar obliquity (spin-orbit misalignment angle) is the main empirical handle.

Rossiter-McLaughlin Effect



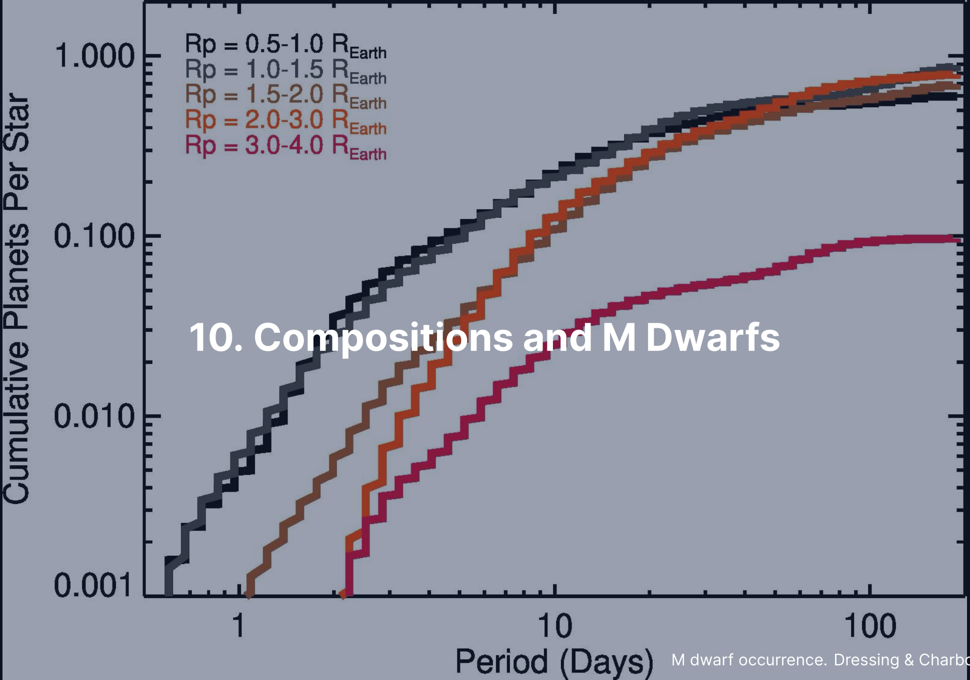
- Transiting planet first occults blueshifted (approaching) hemisphere, then redshifted.
- Time-resolved line distortion traces angle λ between projected stellar spin and orbit normal.
- Triaud (2018) review.

Observed Obliquity Distribution



- λ vs $a/R_\star$, with sub-panels per host spectral type (panel labels too small for back rows; trust the trend, not the tick labels).
- Cool stars ($T_{\rm eff}$ < 6250 K): tight orbits aligned (tidal realignment).
- Hot stars: scattered λ.
- Wide scatter at large $a/R_\star$ implies a wide *primordial* obliquity distribution.
- All three migration channels likely operate.

10. Compositions and M Dwarfs

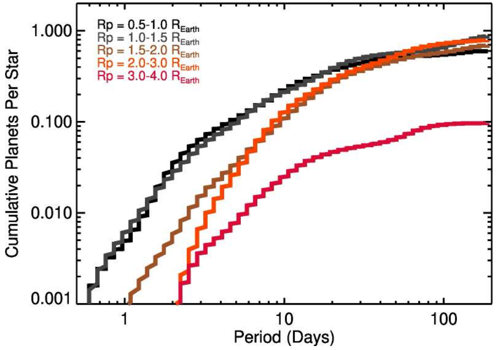


M dwarf occurrence. Dressing & Charbonneau (2015)

Sub-Neptunes: Water World or H/He?

- Super-Earths (below valley): rocky, $\bar{\rho} \sim 4\text{–}8\ \mathrm{g\ cm^{-3}}$.
- Sub-Neptunes (above valley): $\bar{\rho} \sim 1\text{–}3\ \mathrm{g\ cm^{-3}}$, need volatiles on top of a rocky core.
- Volatile component degenerate: thick H/He, thick H_2O, or some mix.
- **Hycean** hypothesis (Madhusudhan+ 2023): shallow liquid ocean under thick H_2 atmosphere. Contested.
- Ultra-short-period rocky: thin atmosphere vs bare rock indistinguishable from density alone.

M Dwarfs: Easy Targets, Tough Habitability

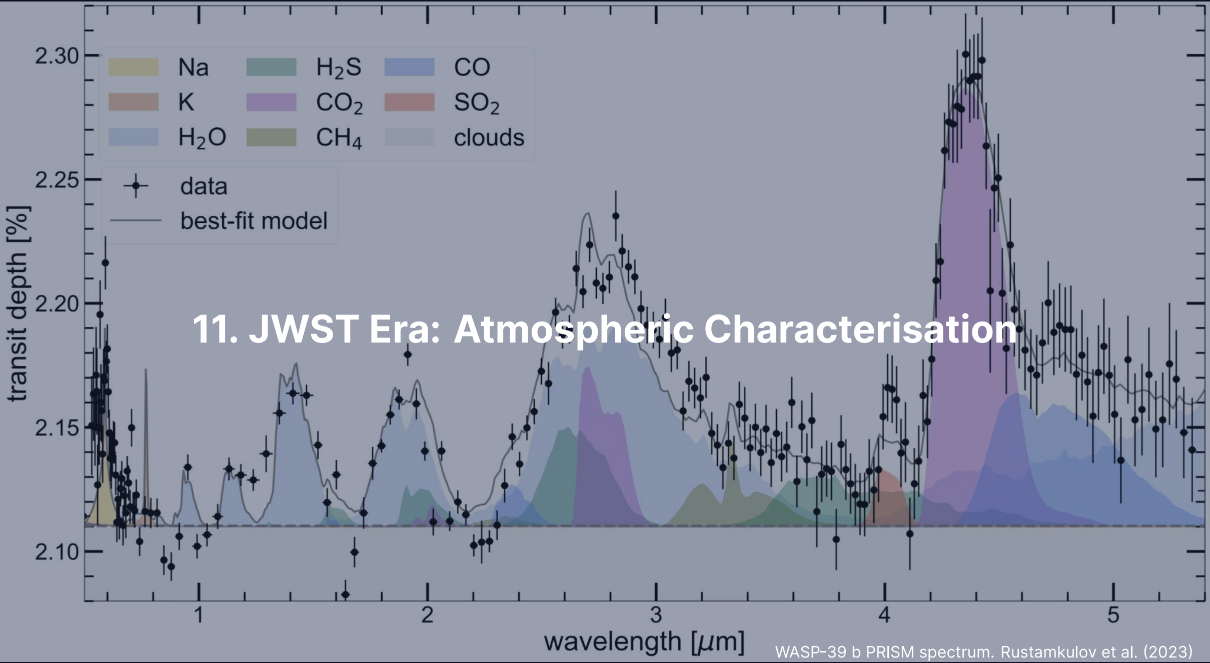


- ~ 75% of all main-sequence stars.
- M dwarfs host ~ 2.5 small planets each; ~ 0.16 in HZ.
- Transit + RV signals geometrically favourable.
- BUT: pre-main-sequence high-luminosity for hundreds of Myr.
- Habitable-zone planet *starts* inside runaway greenhouse zone.
- Dressing & Charbonneau (2015).

M Dwarf Habitability: Two Worries

- **Stellar activity** + early high-XUV → Luger & Barnes (2015) showed an ocean's worth of water can be lost as O_2 (false-positive biosignature: a life-like signal, no biological cause).
- **Tidal locking**: HZ M dwarf planets in 1:1 spin-orbit resonance.
- Permanent dayside / nightside; climate must redistribute heat or condense volatiles on the night side.
- 3D GCMs (Yang+ 2013): substellar clouds can extend the inner edge.
- TRAPPIST-1 = central laboratory for these questions.

11. JWST Era: Atmospheric Characterisation



WASP-39 b PRISM spectrum. Rustamkulov et al. (2023)

Transmission Spectroscopy

Starlight filtered through the atmosphere at transit encodes absorption.

Atmospheric scale height (density e-folding distance):

$$H = \frac{k_\mathrm{B} T}{\mu m_u g}$$

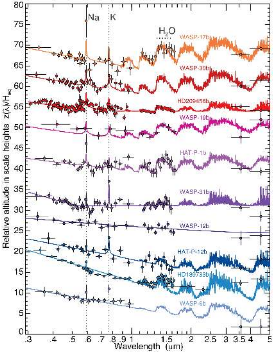
Wavelength-dependent depth: $\delta(\lambda) = (R_p + n_H(\lambda)\, H)^2 / R_\star^2$.

H is wavelength-independent; $n_H(\lambda)$ is the number of scale heights probed (typically a few).

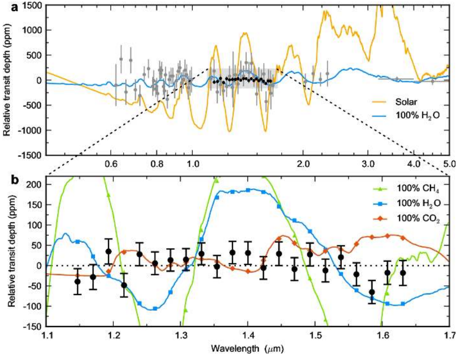
Quick scales for hot Jupiter (T = 1500 K, μ = 2.3, g = 25 m s^{-2}):

- $H \approx 200$ km
- Modulation across ~ 5 scale heights: $\Delta\delta / \delta \approx 2 \times 10^{-2}$
- Absolute modulation: ~200 ppm → JWST-routine.
- Sub-Neptune: 10–100 ppm. Earth/M dwarf: <10 ppm.

Clouds Can Mute Spectra

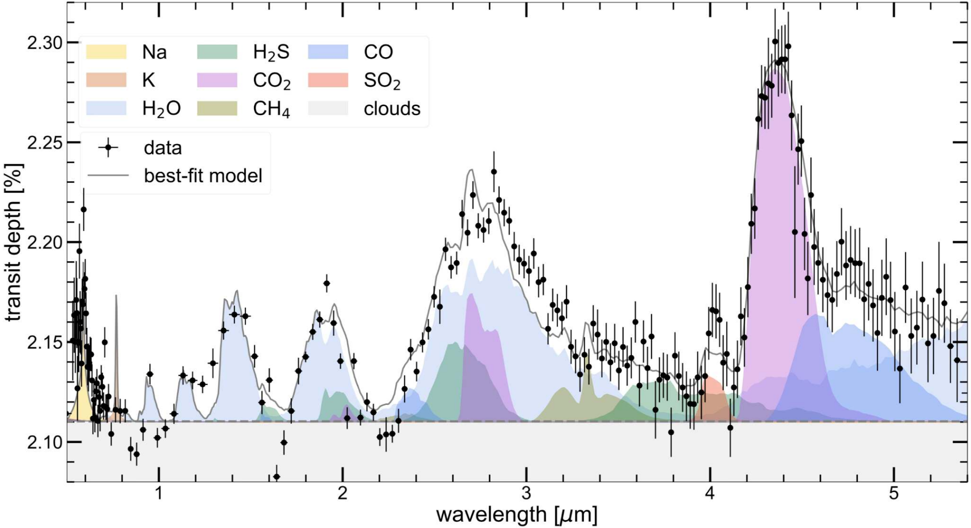


HST + Spitzer, ten hot Jupiters, ordered by optical-to-mid-IR altitude difference. Sing+ 2016.



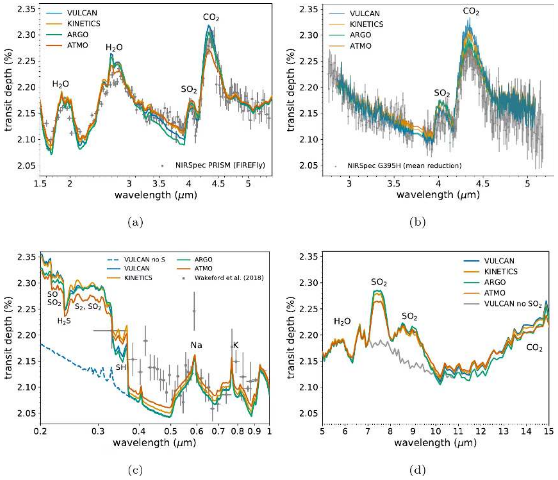
GJ 1214 b: featureless. Kreidberg+ 2014.

Continuum from clear (Na, H_2O at top) to aerosol-dominated (bottom).
Diversity at similar T, g is one of the central puzzles.



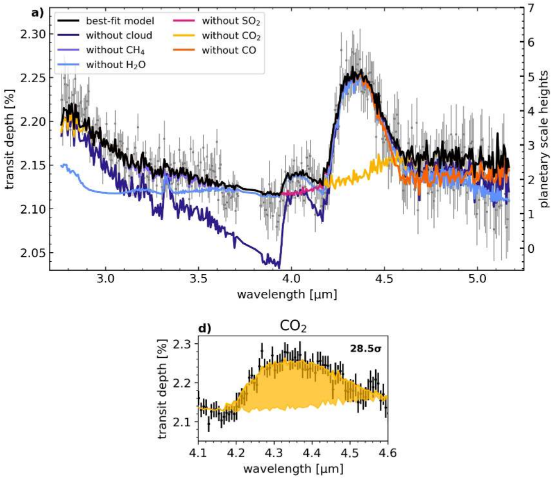
JWST/NIRSpec PRISM transmission spectrum of WASP-39 b. Shaded bands are model opacity contributions; the data support H_2O (33σ), CO_2 (28σ), cloud (21σ), Na (19σ), CO (7σ), not K, H_2S or CH_4. Rustamkulov+ 2023.

WASP-39 b: First Disequilibrium Photochemistry



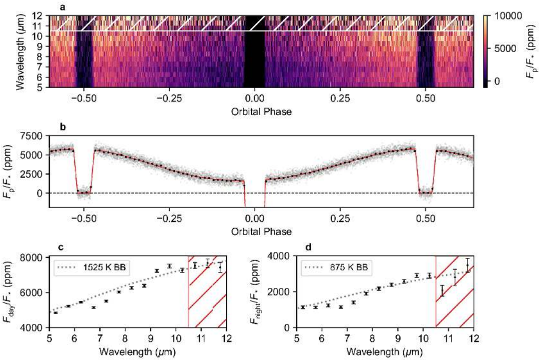
- ERS target: 0.28 M_J hot Saturn.
- The 4.05 μm feature is **photochemical SO_2**.
- Panels are four wavelength ranges; each shows the same four codes (VULCAN, KINETICS, ARGO, ATMO).
- Panel d: MIRI prediction, grey is SO_2 removed.
- SO_2 needs UV photolysis of H_2S, *not* equilibrium chemistry. Tsai+ 2023.

WASP-39 b: 28.5σ CO_2 Feature



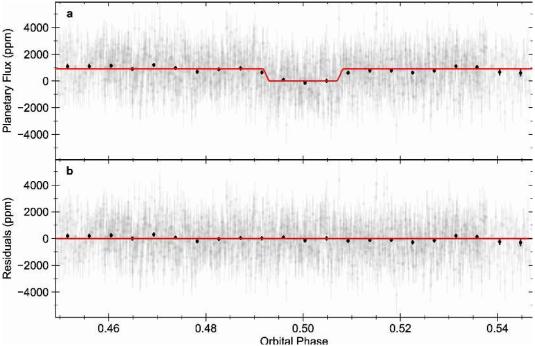
- Each coloured curve: best-fit model with one opacity removed.
- Panel a: full spectrum. Panel d: the CO_2 band alone.
- CO_2 near 4.3 μm: 28.5σ, against 21.5σ (H_2O) and 4.8σ (SO_2).
- Alderson+ 2023.

WASP-43 b: A Phase Curve in Thermal IR

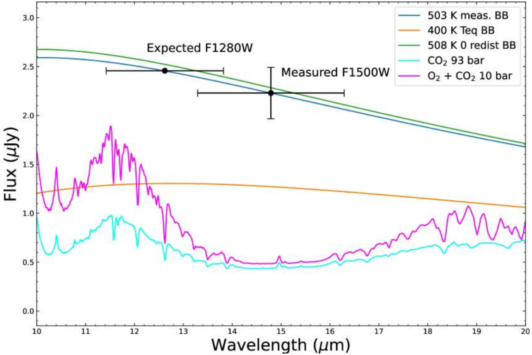


- JWST MIRI phase curve (brightness vs. orbital phase).
- Two transits, two secondary eclipses (planet passes behind the star), smooth phase variation.
- Strong day-night contrast despite thick atmosphere.
- Bell+ 2024.
- Day = hot, night = much colder.

TRAPPIST-1 b: Bare Rock Dayside



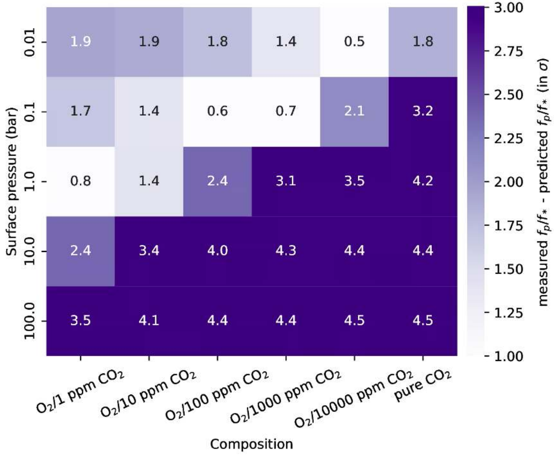
15 μm secondary eclipse



Compared to model atmospheres

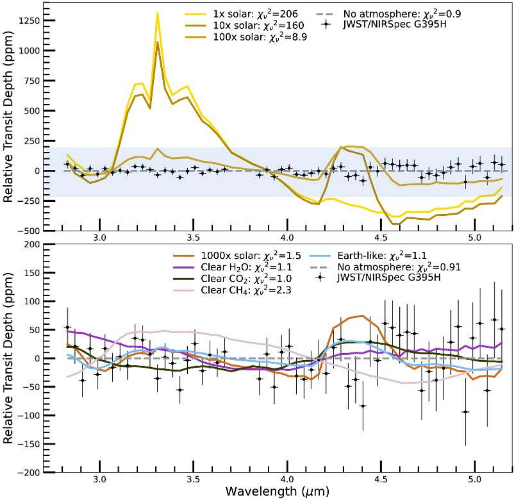
T_B = 503^{+26}_{-27} K, consistent with bare rock equilibrium. **Thick CO_2 atmosphere ruled out.** Greene+ 2023.

TRAPPIST-1 c: Same Story



- Zieba+ 2023.
- Grid of CO_2 partial pressure × thickness.
- ≥ 0.1 bar of CO_2: ruled out.
- Thin atmospheres or none: still possible.
- Inner TRAPPIST-1 planets do not retain Venus-like atmospheres.

LHS 475 b: Featureless

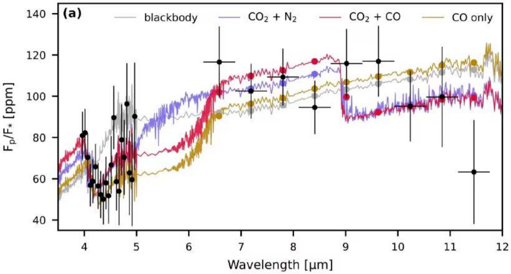


- NIRSpec G395H, Earth-size at 12 pc.
- Flat featureless transmission spectrum.
- H_2/He atmospheres ruled out.
- Pure CH_4 ruled out.
- Pure CO_2 Venus-like marginal.
- Lustig-Yaeger+ 2023.

Pattern: Most Small M Dwarf Planets Lack Atmospheres

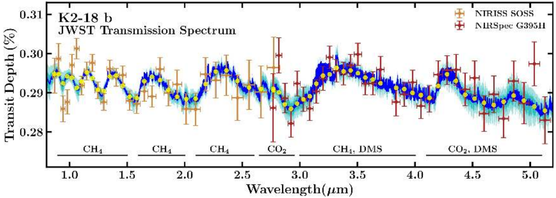
- LHS 475 b, GJ 486 b, GJ 1132 b, TRAPPIST-1 b/c: all consistent with no detectable atmosphere.
- Consistent with Luger & Barnes (2015): early M dwarf XUV strips primordial atmospheres.
- Maybe a watershed: M dwarf HZ planets may not retain water at all.

55 Cancri e: First Atmosphere on a Sun-Like Host?



- Hot rocky super-Earth, ~ 2000 K dayside, plausibly molten.
- NIRCam + MIRI emission.
- Falls below bare-rock blackbody.
- Phase modulation → secondary CO/CO_2 atmosphere outgassed from molten surface.
- Hu+ 2024. *First* tentative atmosphere on rocky planet around Sun-like host.

K2-18 b: Biosignature Controversy

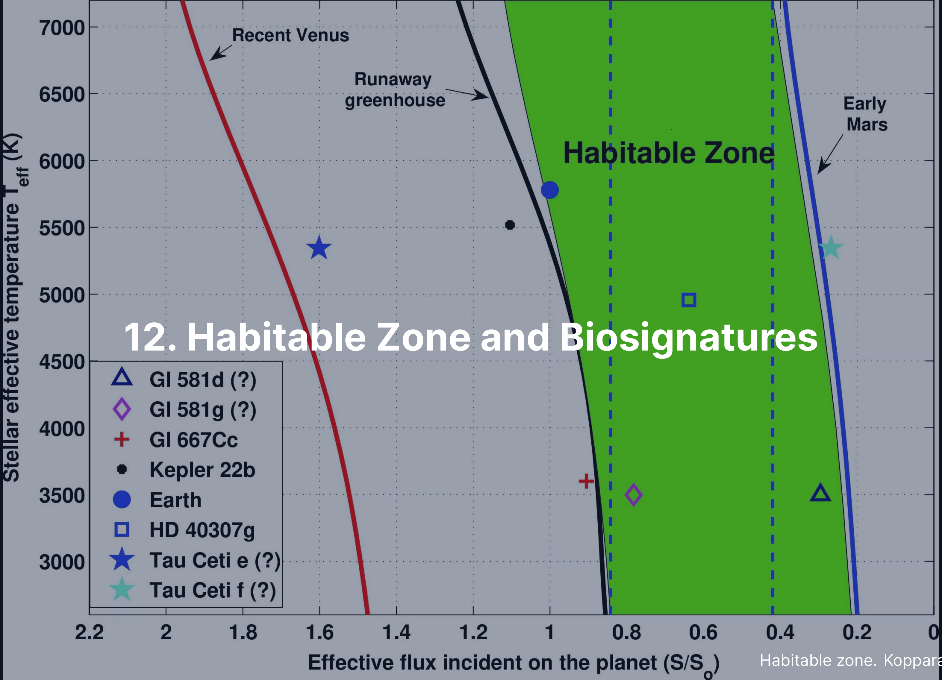


- Sub-Neptune in HZ of M3 host.
- Madhusudhan+ 2023: CH_4 + CO_2 + tentative DMS (~ 3σ).
- Interpreted as **hycean**: H_2 atmosphere over ocean.
- Wogan+ 2024: cannot reproduce DMS; mini-Neptune fit.
- Glein+ 2024: geochemistry against habitable ocean.
- Retrieval-dependent. Debate open.

K2-18 b's Pedagogical Value

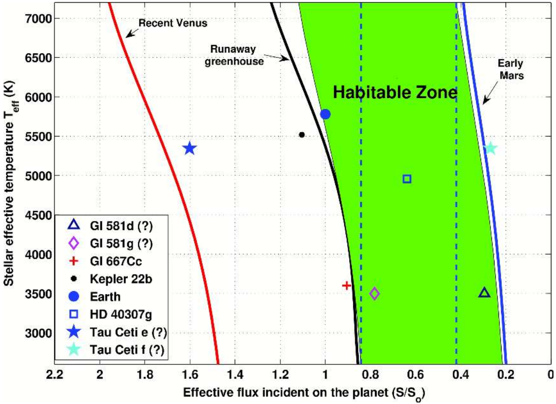
- Whatever the resolution: K2-18 b is the textbook case of how a tentative biosignature is tested in real time.
- Same pattern: ALH84001, Mars methane, Venus phosphine.
- This time the discussion has been faster, more transparent, better documented.
- Useful template for the louder claims of the next decade.

12. Habitable Zone and Biosignatures



Habitable zone. Kopparapu et al. (2013)

The Classical Habitable Zone



- Inner edge: Simpson-Nakajima runaway greenhouse (L9).
- Outer edge: maximum CO_2 greenhouse condenses out as ice.
- Kopparapu+ 2013 modernised Kasting+ 1993.
- Symbols: TRAPPIST-1, Tau Ceti, GJ 581/667, etc.

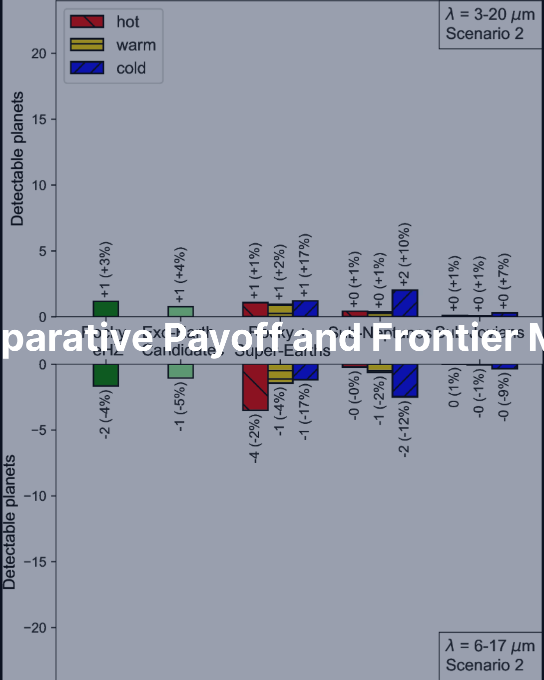
HZ is a Trajectory, Not a Snapshot Line

- History matters more than current location.
- Venus today sits formally inside the HZ if computed naively, yet is hellish (L9).
- M dwarf HZ planets traverse the runaway zone for hundreds of Myr in pre-MS phase.
- 1D Kasting/Kopparapu boundaries shift by 5–20% in 3D GCMs.
- Treat the HZ as a first-order screening tool, not a precise boundary.

Biosignature Gases and False Positives

- Classical: *disequilibrium combinations*. On Earth: O_2 + CH_4.
- Catalogue: O_2, O_3, CH_4, N_2O, organosulphur, organohalogen species.
- No single gas in isolation is convincing.
- **False positives are the main challenge.**
  - Wordsworth+ 2014: H_2O photolysis + H escape → abiotic O_2.
  - CO_2 photolysis → abiotic O_2 via CO + O.
  - Volcanic + serpentinisation → abiotic CH_4.
- Inverse problem: must rule out *all* plausible abiotic pathways.

13. Comparative Payoff and Frontier Missions



LIFE detection yield. Quanz et al. (2022)

Is the Solar System Typical?

Honest answer: **not obviously, but not yet known to be rare**.

Sparsely populated corners:

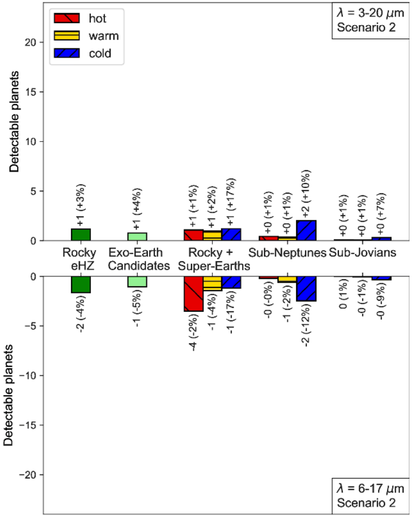
- Most common host: M dwarf, not G dwarf.
- Most common planet: sub-Neptune (2–3 $R_\oplus$). *Solar system has none*.
- Most common architecture: peas-in-a-pod, ~ 5–8 similar planets within 0.2 AU. *Not the inner solar system*.
- No hot Jupiter or hot Neptune in our system.
- Wide low-eccentricity giants (vs RV-discovered eccentric giants).

But: detection biases penalise solar-system analogues. Gaia DR4/DR5 + PLATO will resolve.

Frontier Missions 2026–2035: Transits and Atmospheres

- **PLATO** (ESA, end 2026): Earth analogues in HZ of bright Sun-like stars; asteroseismic stellar radii.
- **Ariel** (ESA, 2029): atmospheric statistics of ~ 1000 exoplanets at 0.5–7.8 μm.
- **Roman** (NASA, 2027): 1000–3000 microlensing planets at 1–10 AU + comparable rogue population; coronagraph tech demo at 10^{-8}–10^{-9}.

Frontier Missions 2030s–2040s: Direct Imaging



- **HWO** (NASA, 2040s): 6 m, 10^{-10} contrast, ~ 25 Earth analogues.
- **LIFE** (ESA concept): nulling interferometer (starlight cancelled by interference), 4–18 μm thermal emission.
- Figure: yield-sensitivity panels from Quanz+ 2022; trends matter, not the small numerical labels.
- **ELT class** (ground, 2028+): 39, 24.5, 30 m. High-contrast + high-resolution NIR.

From Demography to Individual Characterisation

- Today: thousands of detected planets, dozens of well-characterised atmospheres.
- 2040s: full atmospheric retrievals on a curated sample of Earth analogues.
- Open question for L14: what would convince us we have *detected life*?
  - Single biosignature gas?
  - Specific combinations at specific abundances?
  - Temporal variability (seasonal cycles)?
  - Vegetation or photosynthetic surface features?

Summary: Today's Course-Level Takeaways

1. Each detection method picks out a different region of parameter space; the observed archive is the union of biases.
2. Transit + RV → bulk density: the observational machinery that turned exoplanet detections into physical objects.
3. Kepler showed planets are common. $\eta_\oplus \sim 0.3$. Hot Jupiters are rare.
4. **Radius valley** at ~ $1.8\,R_\oplus$: photoevaporation + core-powered mass loss as universal sculptor.
5. Hot Neptune desert + peas-in-a-pod + TRAPPIST-1 chain: three architectural results that any formation theory must explain.
6. JWST: WASP-39 b SO_2, TRAPPIST-1 b/c bare rock, K2-18 b biosignature controversy.
7. Solar system is not obviously typical. Gaia DR4/5 + PLATO will tell us if it is rare.

Questions?

Next: **Lecture 14. Synthesis & Astrobiology**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`